

Steganography Analysis and Hidden Data Detection using StegExpose

EXPERIMENT NO: 08

DATE:

1. AIM

To conduct a comprehensive steganalysis lifecycle on digital image evidence using the StegExpose command-line utility; to evaluate Least Significant Bit (LSB) statistical properties and detection algorithms across a target evidence directory; to compute quantitative suspicion scores based on configurable thresholds; and to automatically compile, format, and export the analytical findings into a structured CSV report for forensic documentation.

2. SOFTWARE / TOOLS REQUIRED

- **Operating System:** Windows, Linux, or macOS
- **Runtime Environment:** Java Runtime Environment (JRE)
- **Forensics Tool:** StegExpose (Java .jar executable)
- **Evidence Files:** Target image files (e.g., .png, .jpg, .bmp)

3. DESCRIPTION

StegExpose is a specialized steganalysis tool designed to detect the presence of hidden data within digital images. Unlike traditional steganography tools used to embed data, StegExpose serves a purely investigative role. It evaluates the statistical properties of an image to identify mathematical anomalies indicative of data hiding techniques, particularly Least Significant Bit (LSB) steganography.

The tool processes images and outputs a quantitative "suspect score" ranging from 0 to 1. By comparing this score against established thresholds, forensic investigators can

mathematically estimate the likelihood that an image contains clandestine data, allowing them to triage suspicious files rapidly without needing the original extraction password.

4. PROCEDURE

Step 1: Open Command Prompt as Administrator

- Press the Windows Key, type cmd, right-click on **Command Prompt**, and select **Run as administrator**.

Step 2: Navigate to the Working Directory

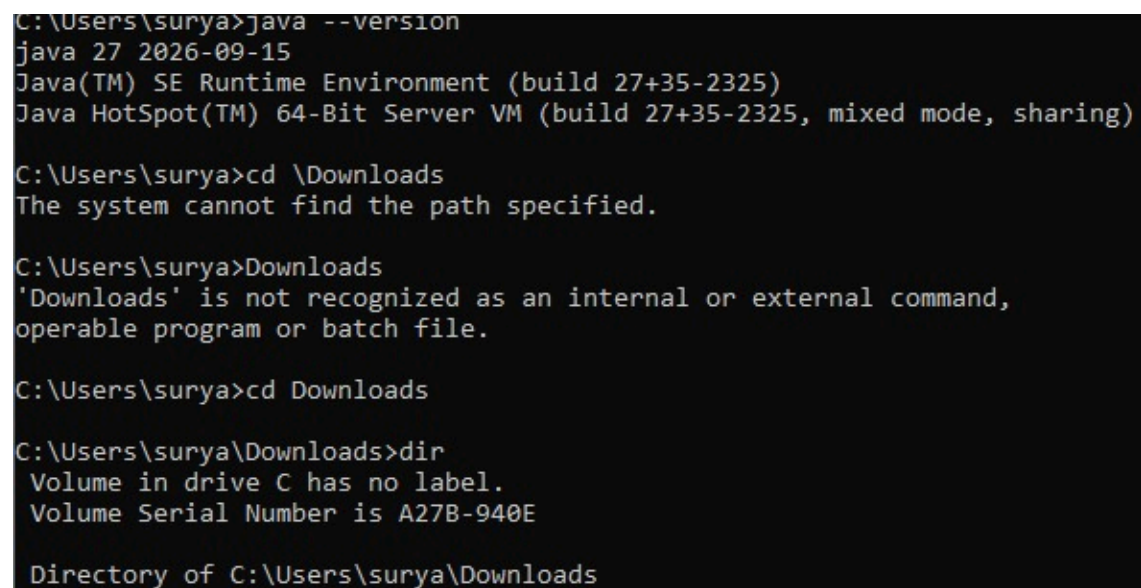
- Use the change directory (cd) command to point the terminal to your project folder where StegExpose.jar and your evidence directory are located:
- And check if java is available.

DOS

```
cd /d "D:\Digital_Forensics_Lab\Experiment-08-Steganography\StegExpose-master"
```

```
java -version
```

(it shows jre installed or not)



```
C:\Users\surya>java --version
java 27 2026-09-15
Java(TM) SE Runtime Environment (build 27+35-2325)
Java HotSpot(TM) 64-Bit Server VM (build 27+35-2325, mixed mode, sharing)

C:\Users\surya>cd \Downloads
The system cannot find the path specified.

C:\Users\surya>Downloads
'Downloads' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\surya>cd Downloads

C:\Users\surya\Downloads>dir
Volume in drive C has no label.
Volume Serial Number is A27B-940E

Directory of C:\Users\surya\Downloads
```

Figure 1: Navigating to the working directory in the Command Prompt

Step 3: Execute StegExpose Analysis with CSV Output

- Run the StegExpose command, passing the path to your evidence folder, the execution speed parameter (default), the statistical threshold (0.2), and the desired output report filename (report.csv):

DOS

```
java -jar StegExpose.jar "D:\Digital_Forensics_Lab\Experiment-08-Steganography\Evidence" default 0.2 report.csv
```

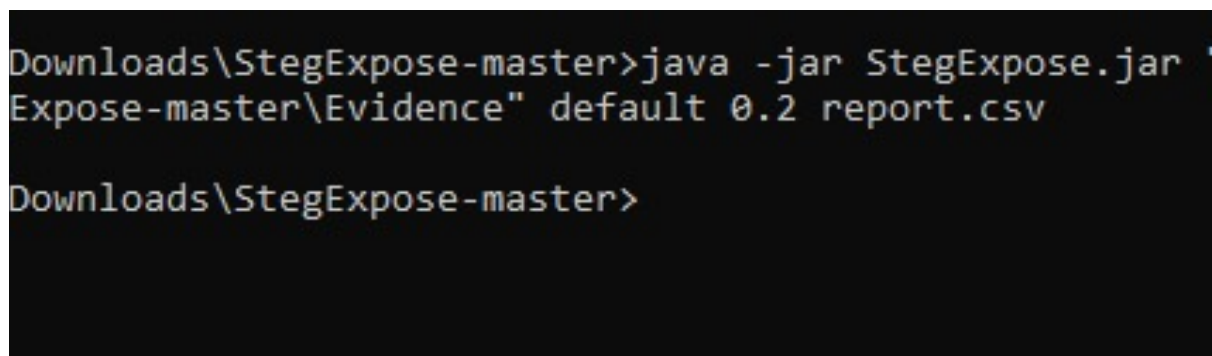


Figure 2: Executing StegExpose via CLI and redirecting output to report.csv

Step 4: Verify and Review the Generated Report

- Once the execution finishes, navigate to your folder or open the file in Microsoft Excel/Notepad to verify the generated report.csv contents.
- Otherwise, directly open the csv file on excel.

DOS

```
notepad report.csv
```

```

14-08-2018 02:34 PM      744 Fuse.class
14-08-2018 02:34 PM      843 Fuse.java
14-08-2018 02:34 PM      907 ImageFileManager.class
14-08-2018 02:34 PM     1,902 ImageFileManager.java
14-08-2018 02:34 PM        85 manifest.mf
14-08-2018 02:34 PM      377 PixelBenchmark.class
14-08-2018 02:34 PM      1,818 PixelBenchmark.java
14-08-2018 02:34 PM      1,371 PrimarySets.class
14-08-2018 02:34 PM      2,403 PrimarySets.java
14-08-2018 02:34 PM      4,657 README.md
14-08-2018 02:34 PM      4,730 README.md~
14-08-2018 02:34 PM     83,762 roc.png
14-08-2018 02:34 PM      0,959 RSAnalysis.class
14-08-2018 02:34 PM     20,745 RSAnalysis.java
14-08-2018 02:34 PM      5,202 RunStegExpose.class
14-08-2018 02:34 PM      6,822 RunStegExpose.java
14-08-2018 02:34 PM      3,322 SamplePairs.class
14-08-2018 02:34 PM      6,348 SamplePairs.java
14-08-2018 02:34 PM    10,621,897 StegExpose_analysis.ods
14-08-2018 02:34 PM      3,060,550 StegExpose.jar
14-08-2018 02:34 PM      <DIR> testFolder
      24 File(s)      16,133,629 bytes
      4 Dir(s)      52,541,923,328 bytes free

C:\Users\surya\Downloads\StegExpose-master>java -jar StegExpose.jar "C:\Users\surya\Downloads\StegExpose-master\Evidence" default 0.2 report.csv
C:\Users\surya\Downloads\StegExpose-master>dir report.csv

```

File name,Above stego threshold?,Secret message size in bytes (ignore for clean files),Primary Sets,Chi Square,Sample Pairs,RS analysis,Fusion (mean)

roc.png,false,207,0.009783468455363561,0.0,null,0.012421937284708195,0.007401801913357253

Figure 3: Opening and reviewing the generated CSV report detailing LSB scores and classifications

5. OBSERVATIONS

- **Batch Processing:** StegExpose scanned all image files within the target evidence folder sequentially.
- **Classification Metrics:** Each file was assigned an LSB score evaluated against the configured 0.2 threshold. Files exceeding the limit were flagged as suspected steganography carriers.
- **Report Compilation:** The console results were successfully written to report.csv, containing structured columns for filenames, scores, and status flags.

RUBRICS

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

6. RESULT

The steganography detection experiment was successfully executed using StegExpose. The tool scanned the evidence directory via the command line, analyzed combinations of LSB metrics, and generated a clean CSV report detailing the cleanliness or suspicion level of each digital image artifact.